

Report R26 — Temperley–Lieb and the four unitary rules

What maps, what does not, and why the two families scale differently

Tier 1 analysis

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Abstract

The four unitary single- V rules $W_{108}, W_{201}, W_{156}, W_{198}$ have been described in this project as mapping to a Temperley–Lieb (TL) model. This report establishes exactly how much of that is true. **One:** the gate is $g_i = \mathbb{K} - 2e_i$ with e_i an exact orthogonal projector, and on the full 2^N space the TL relation fails for all four — what holds instead, exactly, is $e_i e_{i\pm 1} e_i = \tau e_i P^{\text{spec}}$ with a spectator projector two sites away. **Two:** on the *kink module* of a wall-free gap the spectator is pinned and the obstruction disappears: for W_{156}/W_{198} the generators become rank one, the relations hold exactly with arbitrarily many generators, and the algebra is genuinely TL at $\delta = 2\sqrt{2}$, with gap algebra $\mathbb{C} \oplus M_g(\mathbb{C})$ of dimension $g^2 + 1$. **Three:** W_{201}/W_{108} are *not* TL on any space we can find; their gap module is the hard-core chain and their generators have rank growing with the gap. **Four:** that single structural difference — gap module linear in the gap length versus exponential — is the entire origin of the different n_{wcc} and d_{max} scalings, with no singularity at any TL parameter required. **Five:** the fragmentation is quantum, and we exhibit the state that makes it so: a one-dimensional dark line on each gap, the Rokhsar–Kivelson superposition over domain-wall positions with amplitude ratio exactly $\sqrt{2} - 1 = \tan(\pi/8)$, which is not a product state in the site basis. **Six:** the TL–fragmentation literature works at $\delta = m$, an *integer* set by a colour count; our $\delta = 2\sqrt{2}$ is irrational and arises on a *qubit* chain from a Hadamard, so the model is outside the family that has been studied. Finally we record that two premises of the question as posed do not survive: no linear MPO growth has ever been measured for any of these rules, and the bounded-versus-exponential comparison that motivated the question was between dissipative children, not between these unitary parents.

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1 Temperley–Lieb in one page, for a physicist who has not met it

The Temperley–Lieb algebra $TL_n(\delta)$ is generated by $e_1, \dots, e_n$ subject to three relations,

$$e_i^2 = \delta e_i, \quad e_i e_{i\pm 1} e_i = e_i, \quad [e_i, e_j] = 0 \quad (|i - j| \geq 2). \quad (1)$$

The single parameter δ is the *loop parameter*: in the diagrammatic picture the e_i are cup–cap diagrams, multiplication is stacking, and every closed loop produced may be erased at the cost of a factor δ . That is the whole content of the first relation.

Two normalisations circulate and it is worth fixing ours. If one prefers the generators to be *projectors*, put $e_i = E_i/\delta$; then

$$e_i^2 = e_i, \quad e_i e_{i\pm 1} e_i = \tau e_i, \quad \tau = \delta^{-2}. \quad (2)$$

We use (2) throughout, because the objects that arise from our gates are projectors on the nose. So $\delta = 1/\sqrt{\tau}$.

Three facts a newcomer needs.

δ decides everything. $TL_n(\delta)$ is semisimple for generic δ , and its representation theory degenerates at $\delta = 2 \cos(\pi/(m+1))$. The value $\delta = 2$ is the boundary.

Not every δ is reachable by projectors. If the e_i are *orthogonal* projectors on a Hilbert space and (2) holds for arbitrarily many generators, then Jones’ index theorem forces $\delta^2 \in \{4 \cos^2(\pi/m)\}_{m \geq 3} \cup [4, \infty)$. A δ strictly between the discrete values and below 2 is impossible. We use this below to rule out one rule outright.

Why it appears in fragmentation. If the local terms of a Hamiltonian or circuit generate TL, then the conserved quantities are the commutant of TL, which is known in closed form. When that commutant grows exponentially in system size one has Hilbert-space fragmentation, and when its sectors are *not* spanned by product states the fragmentation is called quantum [1, 2].

2 The generators, and the obstruction on the full chain

Each of the four rules applies a Hadamard at site i conditioned on the two neighbours being in one fixed pattern (m^*, n^*) , the *V-context*:

$$W_{201} : (0, 0), \quad W_{108} : (1, 1), \quad W_{156} : (1, 0), \quad W_{198} : (0, 1).$$

Since H is a reflection, $H = \mathbb{K} - 2|h_-\rangle\langle h_-|$ with $|h_-\rangle$ its -1 eigenvector, the controlled gate is

$$g_i = \mathbb{K} - 2e_i, \quad e_i = P_{i-1}^{m^*} \otimes |h_-\rangle\langle h_-|_i \otimes P_{i+1}^{n^*}, \quad (3)$$

and e_i is a product of commuting projectors on disjoint sites, hence an exact orthogonal projector. Measured: $\|e_i^2 - e_i\| = 0$ exactly, and $[e_i, e_j] = 0$ exactly for $|i - j| \geq 2$, all four rules, $N = 5..10$. So two of the three TL relations (2) hold identically.

The third does not.

Proposition 1 (The spectator identity). *For all four rules, exactly,*

$$e_i e_{i+1} e_i = \tau e_i P_{i+2}^{n^*}, \quad e_i e_{i-1} e_i = \tau e_i P_{i-2}^{m^*}, \quad \tau = |\langle m^* | h_- \rangle|^2 |\langle n^* | h_- \rangle|^2.$$

Verified to $\leq 1.4 \times 10^{-17}$ for $N = 5..10$. The middle generator drags in a projector *two sites beyond* the window of e_i , which e_i itself does not carry. On the full 2^N space that spectator is a genuine obstruction and the TL relation fails — residuals 1.07×10^{-1} (W_{156}/W_{198}), 1.83×10^{-2} (W_{201}), 6.22×10^{-1} (W_{108}). It fails on the hard-core constrained sector too.

With $|\langle 0|h_- \rangle|^2 = 1/(4 + 2\sqrt{2})$ and $|\langle 1|h_- \rangle|^2 = (3 + 2\sqrt{2})/(4 + 2\sqrt{2})$ the coefficient is

rule	V-context	τ	$1/\sqrt{\tau}$
W_{108}	(1, 1) symmetric	0.728553	$4 - 2\sqrt{2} = 1.171573$
W_{156}, W_{198}	(1, 0), (0, 1) domain wall	1/8 exactly	$2\sqrt{2} = 2.828427$
W_{201}	(0, 0) symmetric	0.021447	$4 + 2\sqrt{2} = 6.828427$

The asymmetric case picks up one large and one small overlap, and their product is 1/8 exactly; the symmetric cases square a single overlap. Note $(4 - 2\sqrt{2})(4 + 2\sqrt{2}) = 8 = (2\sqrt{2})^2$.

Remark 1. Only the middle row is a TL parameter. For W_{108}/W_{201} the number $4 \mp 2\sqrt{2}$ is the coefficient in Proposition 1 and nothing more, because §4 shows those rules never become TL. It should not be printed in a “ δ ” column, which would imply a structure that is absent.

3 Where the obstruction disappears: the kink module

The spectator identity says TL holds wherever the far site is *already* pinned to the V -context value. That is exactly what happens inside a wall-free gap, and it is why the correct variables are not the site-local ones.

For W_{156} the V -context is (1, 0) and a gap admits no ascent, so its basis states are the descending configurations $1^k 0^{l-k}$. Write $|k\rangle$ for the position of the domain wall. Then g_i is active exactly on $\{|i\rangle, |i+1\rangle\}$ and e_i becomes rank one.

Proposition 2 (TL on the kink chain). *On the kink module of a wall-free gap of W_{156} or W_{198} , each live generator is $e_i = |v_i\rangle\langle v_i|$ with $v_i \propto |i\rangle - (1 + \sqrt{2})|i+1\rangle$, adjacent overlaps $|\langle v_i|v_{i+1}\rangle|^2 = 1/8$ exactly, and the relations (2) hold with $\tau = 1/8$, i.e. $\delta = 2\sqrt{2}$, for every adjacent pair, with the number of generators growing as the gap.*

Verified independently here, by building the module from the site operators (3) and projecting — not by re-implementing the algebra:

N	module dim	live gens	ranks	closure	TL error	$ \langle v_i v_{i+1}\rangle ^2$	ratio
5	6	3	$\{1\}$	0	1.4×10^{-17}	0.125000	-2.414214
8	9	6	$\{1\}$	0	1.4×10^{-17}	0.125000	-2.414214
10	11	8	$\{1\}$	0	1.4×10^{-17}	0.125000	-2.414214

with $-2.414214 = -(1 + \sqrt{2})$ and $0.125 = 1/8$ exactly, and distant generators commuting to 0. The gap algebra is

$$\mathcal{A}_{\text{gap}} \cong \mathbb{C} \oplus M_g(\mathbb{C}), \quad \dim \mathcal{A}_{\text{gap}} = g^2 + 1, \quad (4)$$

measured as 10, 17, 26, 37, 50, 65 for $g = 3, \dots, 8$. Since $\delta = 2\sqrt{2} > 2$ this is the standard semisimple-TL statement, so TL is not decoration here — it is what proves the gap decomposition.

Remark 2 (A trap worth recording). Testing (2) in the site-local variables on 2^N gives “longest TL run = 2” for all four rules, which reads as a decisive negative. It is an artefact: for W_{156}/W_{198} three adjacent generators are never simultaneously live in any sector, so TL_3 is never even testable there. The whole chain is a *product over gaps* of TL algebras with the gap boundaries frozen, not one TL algebra on 2^N . This report’s first draft drew the wrong conclusion from exactly this.

4 W_{201} and W_{108} are not Temperley–Lieb

For the symmetric rules the wall word is 11 or 00, so a wall-free gap is not a kink chain but the *hard-core* chain — configurations with no two adjacent 1s (resp. 0s) — whose dimension is exponential in the gap length. On their own largest block the generators are nowhere near rank one: at $N = 10$, W_{201} at obc0 has module dimension 144 with generator ranks $\{34, 39, 40, 42, 55\}$, and W_{108} module dimension 55 with ranks $\{13, 15, 16, 21\}$. The TL residuals stay at 1.8×10^{-2} and 6.2×10^{-1} .

There are therefore two independent obstructions, and they are different for the two rules:

- W_{108} : Jones’ index. $\delta^2 = (4 - 2\sqrt{2})^2 = 24 - 16\sqrt{2} = 1.3726$ lies in $(1, 2)$, and the nearest admissible discrete value $4\cos^2(\pi/3) = 1$ is far away. No representation by orthogonal projectors exists for arbitrarily many generators, whatever variables are chosen.
- W_{201} : *not* Jones — $\delta = 6.83 > 2$ is perfectly admissible. The obstruction is the direct one: the generators have rank growing with the gap and simply do not satisfy (2).

5 Why the two families scale differently

This answers the question that motivated the report, and the answer needs no singularity.

The minimal wall word is the bit complement of the V -context. That one fact splits the family:

rules	V -context	wall word	gap module	dimension in gap length l
W_{156}, W_{198}	$(1, 0), (0, 1)$	01, 10	kink chain $1^a 0^b$	$l + 1$, linear
W_{201}, W_{108}	$(0, 0), (1, 1)$	11, 00	hard-core chain	exponential

A gap that admits no ascent has exactly one degree of freedom, the position of its domain wall. A gap with a hard-core constraint has a Fibonacci number of states. Largest block dimension, $N = 4..12$: W_{201} at obc0 gives 8, 13, 21, 34, 55, 89, 144, 233, 377 (Fibonacci), W_{108} gives 3, 5, 8, 13, 21, 34, 55, 89, 144, both families at pbc give Lucas numbers (hard core on a ring), while W_{156}/W_{198} give 4, 5, 6, 9, 12, 16, 20, 27, 36, products of kink chains. This is consistent with the $F(N + 2)$ found independently in R24 for the constrained space of W_{201} .

Linear versus exponential gap-module dimension is the whole of it: it fixes n_{wcc} , it fixes d_{max} , and it fixes the sector-size statistics. There is nothing to reconcile *inside* TL because only one family is in TL. And it answers the entanglement question in the same breath — a module of dimension $l + 1$ cannot carry what a Fibonacci module can.

6 The fragmentation is quantum, and here is the state

Fragmentation is called *quantum* when the sectors are not spanned by an orthonormal product basis. For W_{156}/W_{198} the witness is explicit and we exhibit it.

The summand $\mathbb{C}$ in $\mathcal{A}_{\text{gap}} \cong \mathbb{C} \oplus M_g(\mathbb{C})$ is a *dark line*: a state annihilated by every generator of the gap. Solving for the common kernel:

rule	N	kernel dim	amplitude ratio	support (kink states)
W_{156}	5, ..., 10	1	0.414214	$N - 1$
W_{198}	5, ..., 10	1	2.414214	$N - 1$

with $0.414214 = \sqrt{2} - 1 = \tan(\pi/8)$ exactly and 2.414214 its reciprocal, the mirror rule giving the mirror ratio. So the dark state is the Rokhsar–Kivelson superposition

$$|\text{dark}\rangle \propto \sum_k t^k |k\rangle, \quad t = \sqrt{2} - 1 = \tan(\pi/8), \quad (5)$$

over *all* domain-wall positions in the gap. Its support is every kink state of the gap, so in the computational (site) basis it is manifestly not a product state, and it cannot be rotated to one without leaving the sector.

In which basis, explicitly. The fragmentation is classical in the kink coordinate $|k\rangle$ — the sectors are gaps, labelled by wall positions — and quantum in the site basis, because one conserved direction per gap is the entangled superposition above. Equivalently: the sector decomposition is spanned by product states, but admits an *orthonormal* product basis only when every block is a subcube, which a census over the four rules, two boundary conditions and $N = 3..16$ found in exactly three cases, all at $N = 3$.

7 The literature: this model is not the one that has been studied

The TL-fragmentation literature works at integer δ on multi-level chains. Moudgalya and Motrunich [1] define the TL models on an m -level chain by $\hat{e}_{j,j+1} = \sum_{\alpha,\beta=1}^m (|\alpha\alpha\rangle\langle\beta\beta|)_{j,j+1}$ and state the relations with

$$q + q^{-1} = m, \quad \text{i.e.} \quad \delta = m \in \{2, 3, 4, \dots\}.$$

Li, Sala and Pollmann [2], whose open-system TL model is the closest published neighbour to this project, use the spin-1 case $m = 3$: their $e_{j,j+1}$ projects onto the dimer $\frac{1}{\sqrt{3}}(|++\rangle + |00\rangle + |--\rangle)$, so $\delta = 3$. Klocke, Moore and Buchhold [3] use an $SU(N)$ -symmetric TL Hamiltonian, again with δ set by a colour count.

Ours is a **qubit** chain — two levels — at $\delta = 2\sqrt{2} = 2.828\dots$, irrational, arising from the rank-one spectral projector of a Hadamard rather than from any colour multiplicity. On a two-level chain the standard construction gives $\delta = m = 2$, exactly the critical Jones value; we sit strictly above it. The model is therefore *not* a member of the pair-flip family that the fragmentation literature treats, and we have found no paper reporting a TL fragmentation model at irrational δ on qubits, nor one arising as the bond algebra of a Floquet cellular automaton.

Two further recent papers were checked and do not overlap: Zhou, Yang and Chen [4] (quantum HSF and entangled frozen states, TL among four models, entanglement $S \sim \sqrt{L}$) and Han, Hart, Khudorozhkov and Nandkishore [5] (a construction protocol for quantum-HSF Hamiltonians; TL not used).

Remark 3 (The question as posed cannot be asked). Part of the motivating question was whether anyone had previously reported the “linear evolution-operator MPO growth” of the W_{156}/W_{198} model. That measurement does not exist. What has been measured, for rule 156 alone, is an N -independent master curve with increments of exactly 6 per period in an early window, a concave approach thereafter, a departure law $t_{\text{dep}} \approx 0.91N - 8.4$, and an exact plateau $\chi_{\text{sat}} = (N^2 + 15)/4$ — *quadratic* in N , not linear in t . Those are the results worth asking the literature about, and we have found no report of them. There is no MPO data at all for $W_{198}, W_{201}, W_{108}$, so the linear-versus-exponential comparison between the two unitary families has never been run; the bounded-versus-exponential contrast that motivated the question was between the *dissipative children* W_{28}/W_{70} , W_{29}/W_{71} , W_{157}/W_{199} and W_{73}/W_{109} , not between these parents.

8 Summary

1. $g_i = \mathbb{K} - 2e_i$ with e_i an exact orthogonal projector; two of the three TL relations hold identically for all four rules.
2. The third fails on 2^N , and exactly: $e_i e_{i\pm 1} e_i = \tau e_i P^{\text{spec}}$. The spectator is the whole obstruction.
3. On a wall-free gap the spectator is pinned. For W_{156}/W_{198} the generators become rank one and the algebra is genuinely TL at $\delta = 2\sqrt{2}$, with $\mathcal{A}_{\text{gap}} = \mathbb{C} \oplus M_g(\mathbb{C})$, $\dim = g^2 + 1$. The chain object is a product over gaps, not one TL algebra.
4. W_{201}/W_{108} are not TL anywhere: hard-core gap module, high-rank generators; and W_{108} is excluded outright by Jones' index, W_{201} is not.
5. Linear versus exponential gap-module dimension explains the different n_{wcc} , d_{max} and sector statistics with no singularity.
6. The fragmentation is quantum; the witness is a one-dimensional dark RK line per gap with amplitude ratio $\sqrt{2} - 1 = \tan(\pi/8)$.
7. The published TL fragmentation models sit at integer $\delta = m$ on m -level chains; ours is irrational on qubits and is not in that family.

Provenance

The kink-variable construction, the gap-module classification and the $\mathbb{C} \oplus M_g$ decomposition are the Tier-2 session's, from its reports R-T15 and R-T17. Every numerical claim attributed to them above was reproduced here independently, from the site operators (3) and a projection onto the module, without using their code: the rank-one property, the $1/8$ overlaps, the $-(1 + \sqrt{2})$ ratio, the TL residuals, $\dim \mathcal{A} = g^2 + 1$, and the dark-state ratio $\sqrt{2} - 1$. The spectator identity of Proposition 1 and the three τ values are this report's. The MPO facts in §7 are the Schmidt-rank session's and are quoted, not reproduced.

Two errors made in the course of this work are recorded because they are instructive. A site-ordering inconsistency (operators indexed from the most significant bit, sector search from the least) was invisible because reflection maps $W_{156} \leftrightarrow W_{198}$ and fixes W_{201}, W_{108} , so each rule was silently tested against its mirror's sectors. And the Jones argument was first stated as excluding both symmetric rules; it excludes only W_{108} .

References

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